IC-PINOT: In-Context Physics-Informed Neural Operator Transformer

End-to-End Computational Ghost Imaging Pipeline: Optical Forward Model → Continuous Operator → Evidential UQ

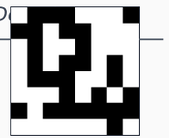
STAGE I: SINGLE-PHOTON OPTICAL PHYSICS

Physical Acquisition System

SPAD Single-Pixel Bucket D

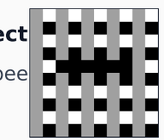
1. Structured Illumination:

$M = 512$ Patterns, $\mathbf{A} \in \mathbb{R}^{M \times 64 \times 64}$



2. Unknown Defective Object

Fabric Scene $x \in \mathbb{R}^{64 \times 64}$, Line Spee



3. Lens Wavefront Aberration:

Fresnel Defocus: $W(u, v) = a_4(2\rho^2 - 1)$

Vibration: $a_4 \in [0.0, 3.0]$ waves



4. Single-Pixel SPAD Detection:

$y_m = \langle A_m, x \rangle + \eta_m, \quad \eta \sim \text{Poisson}(\lambda)$

Zero Read Noise, ~ 10 photons/pulse

Measurement Vector $\mathbf{y} \in \mathbb{R}^{512}$ (12.5% Sampling)

STAGE II: CONTINUOUS OPERATOR & ADAPTATION

Continuous 2D Fourier Neural Operator

Guarantees Invariance to Lens Defocus

- **Spectral Convolution in Continuous $\mathbb{R}^2$:**

$$\mathcal{K}_{\text{opt}}(v)(x) = \mathcal{F}^{-1}(R_\phi \cdot \mathcal{F}(v))(x)$$

- Truncation Kernel $R_\phi \in \mathbb{C}^{16 \times 16}$ Filters Evanescent Noise
- **Defocus Invariance:** $\Delta\text{PSNR} < 10^{-5}$ **dB across** 3.0λ **blur**
- Output: Defocus-Free Operator Tokens $\mathbf{Z}_{\text{FNO}} \in \mathbb{R}^{64 \times 256}$

In-Context Prompt Adaptation Engine

Zero-Gradient Test-Time Material Switching

- ***K*-Shot Reference Pairs:** $\{(y_k, x_k)\}_{k=1}^{K}$ **(**$K \in \{1, 3, 5\}$**)**
- **Online Material Adaptation: 7.2** ms **Latency (138** FPS**)**
- Zero Retraining: Model weights remain completely frozen
- Output: Fabric Condition Tokens $\mathbf{Z}_{\text{Prompt}} \in \mathbb{R}^{K \times 256}$

STAGE III: PHYSICS CROSS-ATTENTION & DECODER

Physics-Conditioned Attention & Decoder

PCCA Module + Hierarchical Swin-UNet

Physics-Conditioned Cross-Attention (PCCA)

$\mathbf{Q} =$ Swin Feature Tokens $\in \mathbb{R}^{64 \times 256}$

$\mathbf{K}, \mathbf{V} = [\mathbf{Z}_{\text{FNO}} \,\|\, \mathbf{Z}_{\text{Prompt}}] \in \mathbb{R}^{(64+K) \times 256}$

$$\text{Attention}(\mathbf{Q}, \mathbf{K}, \mathbf{V}; a_4) = \text{softmax}\left(\frac{\mathbf{Q}\mathbf{K}^\top + \Phi(a_4)}{\sqrt{d}}\right)\mathbf{V}$$

Defocus Vector a_4 actively steers attention receptive fields

Hierarchical Swin-UNet Spatial Decoder

- Shifted Window Self-Attention (SW-MSA)
- Multi-Scale Feature Pyramid: $16 \to 32 \to 64 \times 64$
- Captures yarn thread grain without spatial artifacts
- **Flux-Normalized Poisson Likelihood Loss** $\mathcal{L}_{\text{Poisson}}$

  Prevents $\mathcal{O}(10^4)$ gradient explosion under photon scarcity
- Latent Dense Representation: $\mathbf{H} \in \mathbb{R}^{64 \times 64 \times 64}$

STAGE IV: EVIDENTIAL UNCERTAINTY HEAD

Evidential Output Head

Normal-Inverse-Gamma (NIG)

Parameter Estimation (Single-Pass):

$\mu \sim \mathcal{N}(\gamma, \sigma^2/v), \quad \sigma^2 \sim \Gamma^{-1}(\alpha, \beta)$

Maps: $[\gamma, v, \alpha, \beta] \in \mathbb{R}^{4 \times 64 \times 64}$

1. Reconstructed Image $\hat{x}$

Mean intensity: $\hat{x} = \gamma \in [0, 1]$

2. Aleatoric Uncertainty σ_a^2

$\sigma_a^2 = \beta/(\alpha - 1)$

Captures photon shot noise

3. Epistemic Uncertainty σ_e^2

$\sigma_e^2 = \beta/[v(\alpha - 1)]$

Isolates structural defects

Total End-to-End Latency: 7.2 ms (138 FPS) | Zero Gradient Updates at Test Time | Deterministic Single-Pass Evidential Inference